RDDs Response Xiyao Long.

A Fault Tolerant Abstraction for in memory cluster computing Resilient Distributed Database

RDD system is a distributed data storage and query scheme/system. It decides how to partition large amount of data across many machines based on the keys of the records, therefore it's easy to merge the records into a single database that is hash-mapped in the same way. There are mainly two characteristics of this RDD system. one is that its batch/bulk operation on data in the same way across all distributed locations enable easy and linear logging of the operation. It's easy to recover a. lost partition with the information of RDD operations and other RDDs. Another characteristic is that it can persist the data resulting from certain user queries on to the memory of each distributed location, therefore greatly speed up any ensuing operations such as count/collect/request.

One restriction of RDDS is that its persistence creation and operation can't be done in a per-memory-location basis. It suits for operations that are done in bulk but not suitable if the operation/writes needs to be administered in an arbitrary location. The paper also talked about implementing a distributed machine learning algorithm logistic regression. The system enables us to load a large amount of training data into RAM for gradient calculation. However, I think this advantage may not applicable to many deep learning models because they generally use mini-batch for gradient calculation and parameter/weights updates. It's not obvious how to adapt the mini-batch optimized training strategy into the RDDs scheme unless we know a framework that calculate gradients on different machine but somehow the distributed system can incorporate these calculations across machine to generate a more efficient training process/model.

Besides the above-mentioned point, their implemented system Spark has many features that make the system adaptable and user-friendly. It offers different options for RDDs storage as either in memory or on-disk. It also enables compilation of each user-input line as a class and invoke it in the JVM to give results to user on a per-line basis.